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RESEARCH

Using forced-motion tests to predict nonlinear flutter responses of sectional bridge model via complexification-averaging method

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Abstract
Nonlinear flutter response estimation is important for structural safety and serviceability design. Traditionally, the free vibration test is used to identify the parameters of the fluid-structure interaction, but the corresponding frequencies are very limited. The forced-motion apparatus can measure the aerodynamic force at different frequencies, but can only provide the aerodynamic force and cannot predict the response directly. This study proposes a method to predict the nonlinear flutter response through the aerodynamic force measured by forced-motion equipment. At first, a nonlinear aerodynamic model combines the traditional flutter derivatives for the linear part and the additional polynomials for the nonlinear part, which is identified and fitted by the aerodynamic force autonomous inference method. In the second part, the differential equations of the flutter amplitudes and the phase difference are derived by complexification-averaging methods. The flutter amplitudes at the stable limit-cycle oscillation stage can be calculated when the differential equation equals zero. The amplitude ratio and the phase difference between vertical and torsional motion can also be calculated by the proposed methods.

Keywords Long span bridge · Nonlinear flutter · Nonlinear aerodynamic · Complexification-averaging methods

1 Introduction

Flutter is the most important type of wind-induced vibration for long-span bridges. Since the collapse of the Narrow Tacoma Bridge [1], the flutter instability is the determinant factor in the design of the structures and sectional shape of the long span bridge. In aerospace engineering, flutter vibration is characterized as a coupled pitch (torsion) and plunge (vertical bending) motion of air-foil, and aeroelasticity during flutter can be modeled using potential flow theory [2]. Later, Scanlan developed the flutter derivatives for long-span bridges based on Theodorsen theory [3]. Thereafter, the critical wind speed, at which the flutter occurs onsite, can be evaluated using flutter derivatives. The vibration frequency, the amplitude ratio, and the phase difference between torsional and vertical vibration are also calculated simultaneously.

In contrast to aircraft wings, the dynamic modal frequencies of long-span bridges are typically closer to each other. It is necessary to investigate the flutter state among multiple modes, and even full modes [4]. Normally, the structural dynamic equation and aeroelastic equation are rewritten together in complex notation, then, the critical wind speed can be determined by the complex eigenvalue analysis. Recently, an explicit total damping formula was developed for multi-mode flutter evaluation [5], and other flutter states such as vibration frequency and amplitude ratio are also derived at the same time.

The aeroelasticity in all above flutter analysis is based on Scanlan's flutter derivatives when the wind speeds are close to the critical point. After critical wind speeds, the flutter vibration grows exponentially. However, the vibration amplitudes cannot reach infinity because the energy extracted from the air flow is finite. Therefore, the aeroelasticity inevitably enters into a nonlinear stage as the amplitudes grow, and the

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linear flutter derivatives are no longer valid. Experimental measurements and modeling of nonlinear aeroelastic force are one of the most important research topics in the past decade [6].

Initially, many nonlinear aeroelastic vibration phenomena were observed during sectional tests of the bridge deck in the wind tunnel. Gao et al. [7] reports several nonlinear aeroelastic vibration phenomena. After the critical wind speeds, the initial vibration from equilibrium can still be simulated by linear flutter derivatives. However, after the vibration amplitudes increase, the increasing rate becomes slow to zero. When the amplitudes increasing rate is insignificant, the flutter vibration enters the limit cycle oscillation [8]. For the wind speeds slightly higher than the critical wind speeds, multi-stability was observed [9]. It should be noted that, for wind-induced galloping of iced transmission lines also has similar nonlinear phenomena [10–12].

To simulate the above nonlinear vibration phenomenon, many nonlinear aerodynamic force models were developed from different points of view. The most widely used method is based on the free vibration tests on the sectional model. From the vibration signals measured from the wind tunnel tests, the coefficients of the aerodynamic model can be identified through least squares regression [13]. Another popular method is the direct measurement of the aerodynamic force during flutter vibration [14]. However, the limitation of free vibration is that only the aerodynamics in a very narrow frequency band, which is characterized by the suspension system of the section model, can be measured. In contrast, the forced motion test is also commonly used with adjustable amplitudes and vibration frequency. Normally, the forced-motion test must decouple the vertical and torsional motion [15], therefore, the superposition assumption is required for forced-motion methods [16].

Unlike the straightforward expression of linear aerodynamics as flutter derivatives, the nonlinear aerodynamics is much more complex. To maintain consistency with the linear flutter derivatives, many nonlinear aerodynamic models used the amplitude-dependent flutter derivatives [17]. The fundamental logic of amplitude-dependent flutter derivatives is that, for each vibration cycle, the aeroelastic force is calculated by the derivatives of the corresponding amplitudes for this cycle. The benefit of amplitude-dependent flutter derivatives is compatibility with traditional linear flutter derivatives, such as full-mode aeroelastic simulation based on the finite element method [18].

Nevertheless, the dynamic equation of amplitude-dependent flutter derivatives is not closed-form. Cycle-by-cycle simulation is easy to implement, but global stability and bifurcation analysis are more difficult or impossible. Therefore, the explicit expression of the nonlinear aerodynamic force is developed based on the the bi-variate (displacement and velocity) Taylor expansion, which is used to simulate

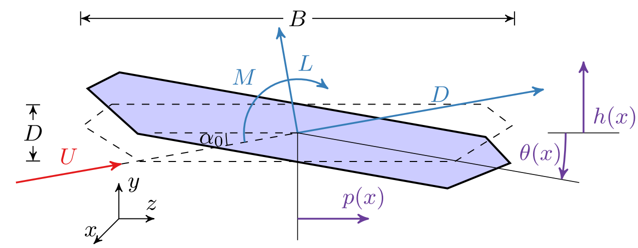


Fig. 1 Wind loads axis and bridge motion axis on bridge section

the force induced by nonlinear vortex [19]. However, a high-order expansion will cause the curse of dimensionality, and there are too many redundant terms during the Taylor expansion of bi-variate polynomial. Traditionally, effective terms selection is empirically based, so it is difficult to achieve a consensus about the nonlinear aerodynamic form. Recently, an autonomous inference method has been developed based on the sparse identification of nonlinear dynamics (SINDy) [20] using free vibration [21] and forced-motion [22] methods. The nonlinear aerodynamic formula is derived autonomously without subjective bias.

The above-mentioned amplitude-dependent flutter derivatives or polynomial expansion-based formula can only provide cycle-by-cycle numerical simulation, which normally employ the Runge-Kutta algorithm [23] to perform time-domain calculation. The analytical formula to directly analyze the nonlinear flutter state from forced-motion measured aerodynamics data is still absent. Furthermore, it is also difficult to perform the global stability and bifurcation analysis. This paper will provide an analytical method to directly predict the bridge nonlinear flutter responses from forced vibration data without traditional free vibration tests.

This paper first provides an aerodynamic modeling through a forced-motion test and a concise formula for nonlinear aerodynamics covering a wide range of frequencies and amplitudes. The superposition assumption will be verified by comparing the aerodynamics between coupled motion and separated motion. In the second part, the nonlinear flutter amplitudes evolution formula will be estimated by using the complexification-averaging (CxA) method. On the basis of the amplitude evolution formula, the flutter response can be estimated from initial instability to stable amplitudes.

2 Aerodynamic modelling through forced-motion tests

2.1 Governing equation of coupled flutter of bridge deck

Aerodynamic loads and bridge motion coordinates are defined in Figure 1. When the bridge section moves sinusoidally in vertical and torsional direction in wind flow with speed as U, the self-excited aerodynamic force is expressed as:

$$L_{se} = L_{se}(h', h, \theta', \theta'; U) \tag{1a}$$

$$M_{se} = M_{se}(h', h, \theta', \theta'; U) \tag{1b}$$

where h, θ are the vertical and torsional vibration amplitudes, respectively; L_{se} and M_{se} are structural motion dependent aeroelastic for certain wind speeds U, and $'$ denote the derivatives respective with time t.

In order to keep consistence with flutter derivatives, the aerodynamic force is expressed as the summation of linear part and nonlinear part as:

$$L_{\text{se}} = \rho U^2 B \left(K H_1^* \frac{h'}{U} + K H_2^* \frac{B\theta'}{U} + K^2 H_3^* \theta + K^2 H_4^* \frac{h}{B} + N_L(h, h', \theta, \theta') \right) \tag{2a}$$

$$M_{\text{se}} = \rho U^2 B^2 \left(K A_1^* \frac{h'}{U} + K A_2^* \frac{B\theta'}{U} + K^2 A_3^* \theta + K^2 A_4^* \frac{h}{B} + N_M(h, h', \theta, \theta') \right) \tag{2b}$$

where ρ is air density; $K = \omega B/U$ is the reduced frequency; B is the bridge deck width; ω is the circular frequency; H_i^* and A_i^* $(i = 1 \sim 4)$ are frequency dependent flutter derivatives, N_L and N_M are nonlinear parts of aerodynamic force for vertical and torsional directions.

Then, the governing equation of coupled flutter is expressed as

$$mh'' + c_h h' + k_h h = \rho U^2 B \left[K H_1^* \frac{h'}{U} + K H_2^* \frac{B\theta'}{U} + K^2 H_3^* \theta + K^2 H_4^* \frac{h}{B} + N_L(h, h', \theta, \theta') \right] \tag{3a}$$

$$I\theta'' + c_\theta \theta' + k_\theta \theta = \rho U^2 B^2 \left[K A_1^* \frac{h'}{U} + K A_2^* \frac{B\theta'}{U} + K^2 A_3^* \theta + K^2 A_4^* \frac{h}{B} + N_M(h, h', \theta, \theta') \right] \tag{3b}$$

The above coupled flutter formula can be converted to dimensionless format through dimensionless displacement $y = h/B$, and dimensionless time $\tau = tU/B$

$$\ddot{y} + 2\xi_y K_y \dot{y} + K_y^2 y = m_y^* \left[K H_1^* \dot{y} + K H_2^* \dot{\theta} + K^2 H_3^* \theta + K^2 H_4^* y + N_L(y, \dot{y}, \theta, \dot{\theta}) \right] \tag{4a}$$

$$\ddot{\theta} + 2\xi_\theta K_\theta \dot{\theta} + K_\theta^2 \theta = m_\theta^* \left[K A_1^* \dot{y} + K A_2^* \dot{\theta} + K^2 A_3^* \theta + K^2 A_4^* y + N_M(y, \dot{y}, \theta, \dot{\theta}) \right] \tag{4b}$$

where ξ_y, K_y, ξ_θ, K_θ are damping coefficients and reduced structural frequencies for vertical and torsional motion, $\dot{}$ indicates the differentiation respect with dimensionless time τ and m_y^* and m_θ^* are mass ratios for vertical and torsional motion.

2.2 Aerodynamic measurement from forced-motion equipment

In order to measure the nonlinear aerodynamic force of the bridge deck subject to sinusoidal motion, the bridge deck is placed on the forced-motion equipment, which is shown in Figure 2. This device can carry the bridge deck undergoing any prescribed motion and measure the aerodynamic force simultaneously. The bridge deck used in this study is a typical streamlined section. The geometry of the full scale and the model scale in the wind tunnel are shown in Figure 3, and geometry scale is 1:100.

The test configuration is listed as in Table 1. It should be noted that vertical and torsional motion-induced aerodynamics are measured separately, and the underlying superposition assumption will be validated in next subsection. The vibration frequency is kept constantly as 3Hz to ensure that possible flutter wind speeds can match with the optimal wind tunnel speeds, which was operated in TJ-7 wind tunnel [15]. Two force balance [25] were placed at two ends of sectional model to measure the lift and torque force induced by bridge motion simultaneously. Because the force balances actually measure the aerodynamic and inertial force of deck model at the same time, the model fabrication prefer light materiels to ensure the force measurement accuracy. For both the vertical and torsional directions, each wind speed is tested with different amplitudes, so there are total configurations $17 \times 9 + 17 \times 8 = 289$.

Figures 4 and 5 show two examples of the aerodynamic force of the bridge deck in torsional or vertical sinusoidal oscillation when wind speed is $U = 7.8$ m/s. Figure 4(a) plots the lift and torque forces induced by torsional vibration when the amplitude is 3°, in which the left part is the aerodynamic force varies periodically with time and the right part is the force varies with displacements. Figure 4b plots the same force when the amplitude is 13°.

If the aerodynamic force varies linearly with the motion of the bridge deck, the enclosed curve of aerodynamic force and displacement should be an ellipse. The curves in Figure 4(a) are more identical to the ellipse, which shows that nonlinearity is not obvious when the amplitude of θ is 3°. When the amplitude of θ is 13°, the curve shape is different. In other words, the nonlinear aerodynamics has a greater contribution than small amplitudes.

The aerodynamic force caused by the vertical motion when the amplitude is 13 mm (the largest) is plotted in Figure 5. In contrast, the force-displacement curve is still

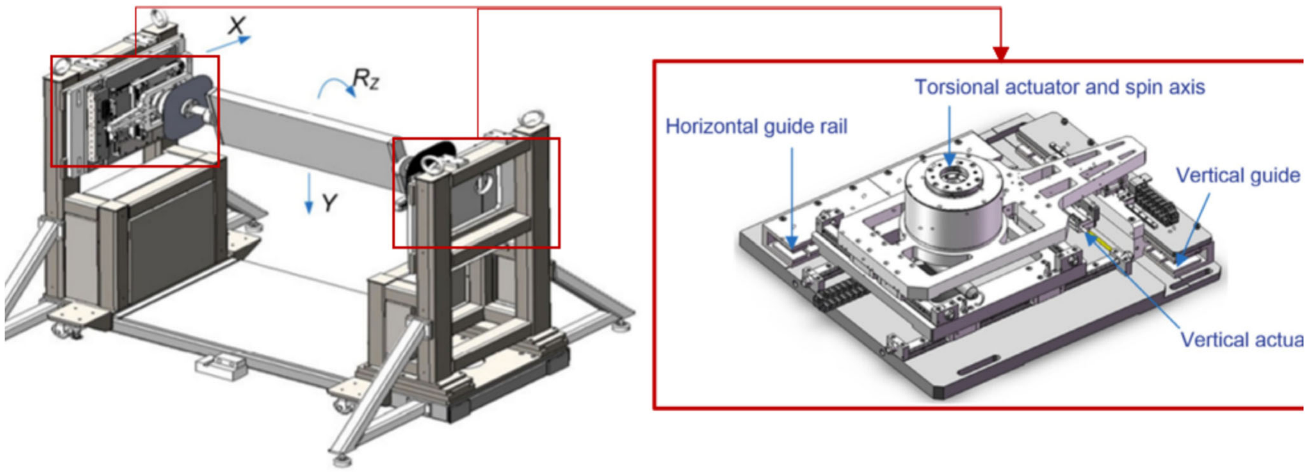


Fig. 2 Schematic diagram of forced-motion device

Fig. 3 Geometry of Sectional model (left: model scale units: mm, right: full scale, units: m; geometries of accessory components can be found in Ge et al. [24])

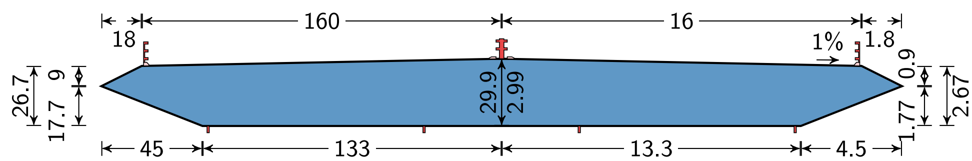


Table 1 Separated forced-motion tests configurations

Motion items	Configurations	Unit
Vibration frequency	3	Hz
Wind speeds	5, 5.5, 6, 6.5, 7, 7.2, 7.4, 7.5, 7.6, 7.8, 8, 8.2, 8.4, 8.5, 9.0, 9.5, 10	m/s
Reduced frequency($\omega B/U$)	1.342, 1.220, 1.118, 1.032, 0.959, 0.932, 0.907, 0.895, 0.883, 0.860, 0.839, 0.818, 0.799, 0.790, 0.746, 0.706, 0.671	
Torsional Amplitudes	1,2,3,4,5,6,7,8,9	°
Vertical Amplitudes	1,2,3,5,7,9,11,13	mm

identical to the ellipse curve for a large vertical amplitude, and the nonlinear aerodynamics is negligible.

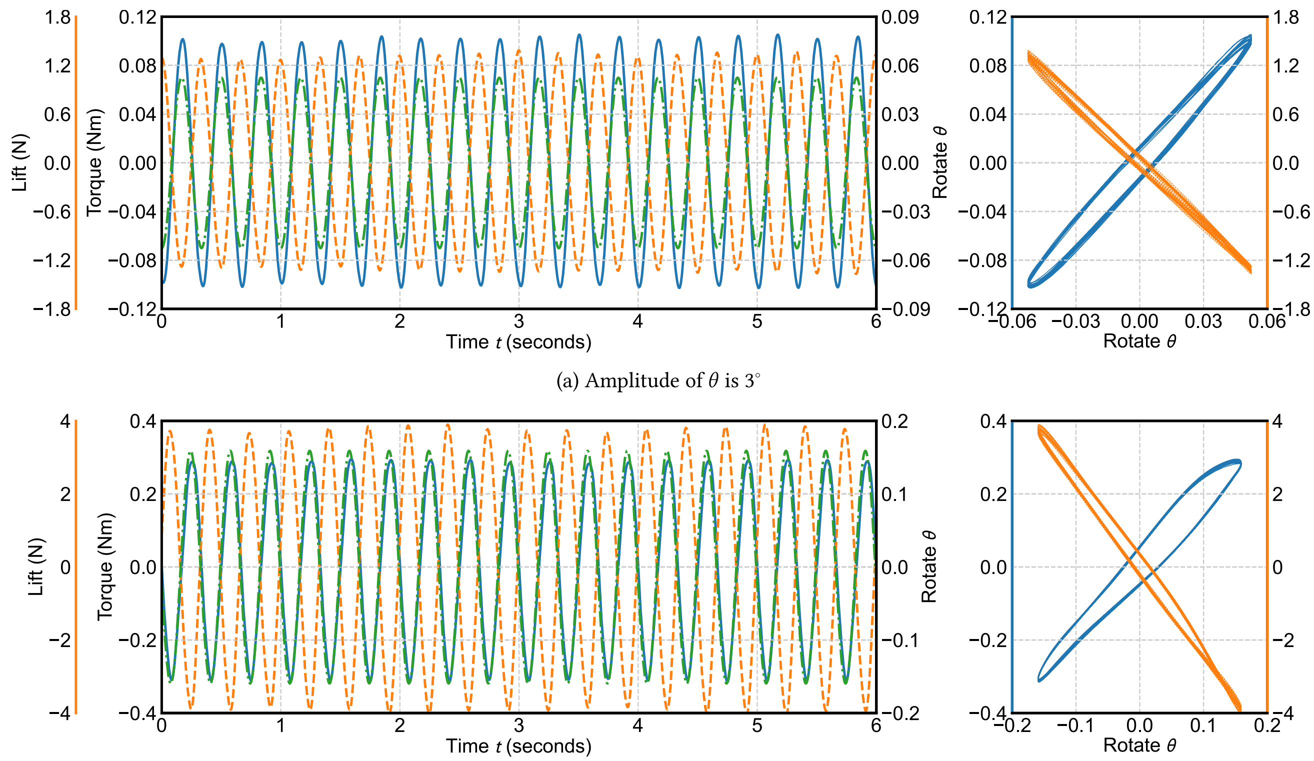
2.3 Validation of superposition assumption

Superposition means the aerodynamics of bridge deck in coupled vertical-torsional motion can be decomposed into vertical motion and torsional motion induced aerodynamics separately. Taken the aerodynamic torque as example, the nonlinear aeroelastic force N_M is an unknown nonlinear formula for position θ, y and velocity $\dot{\theta}$, $\dot{y}$. For such nonlinear fluid-structure interaction, it is difficult to determine the analytical expression because of the complex expression among four vibration states θ, y, $\dot{\theta}$ and $\dot{y}$ and huge number experimental configurations between interaction of vertical and torsional motion. Based on the superposition assumption, N_M can be decomposed as

$$N_M(y, \dot{y}, \theta, \dot{\theta}) = N_{yM}(y, \dot{y}) + N_{\theta M}(\theta, \dot{\theta}) \tag{5}$$

Therefore, the expression of N_M is simplified and the experimental configurations is greatly reduced.

Based on superposition assumption, the aerodynamics can be measured in purely vertical or torsional motion, and then, when the bridge deck is in coupled motion, the aerodynamics is calculated by simple summation of two motion-induced forces together. If the aerodynamic is linear, the superposition is naturally valid because the flutter derivatives only have first-order items. However, if the aerodynamic is nonlinear, the superposition assumption is questionable. If the superposition assumption is not valid, the nonlinear aerodynamic will be very complex, because there are too many combi-



(a) Amplitude of θ is 3°

(b) Amplitude of θ is 13°

Fig. 4 Nonlinear Aerodynamics for the bridge deck in torsional sinusoidal oscillation when $U = 7.8$ m/s

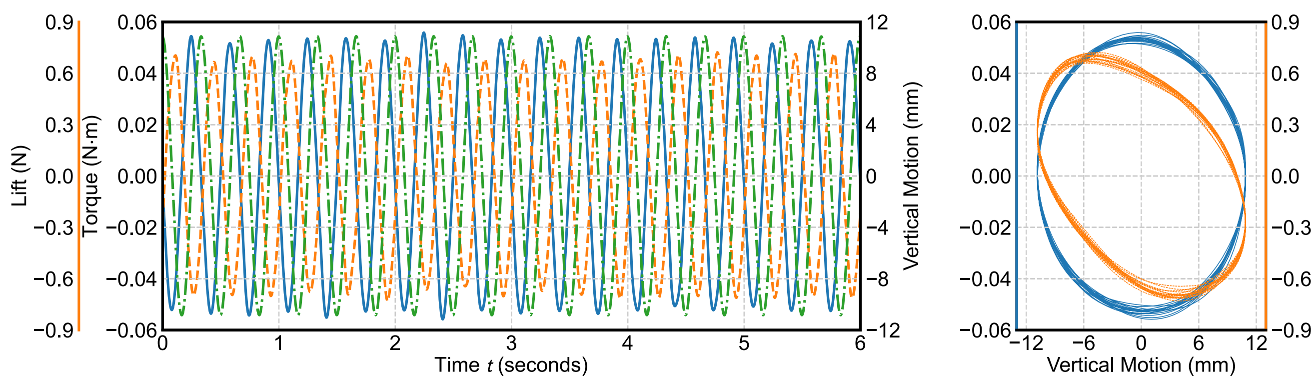


Fig. 5 Linear Aerodynamics for the bridge deck in vertical sinusoidal oscillation

nations of amplitudes and phase difference of vertical and torsional motion.

Based on the forced-motion device, the aerodynamics in coupled motion can be compared with the superposition of measured aerodynamic from purely vertical and torsional motion. Figure 6 plots the aerodynamic comparison between coupled motion and superposed from the results of Table 1 when the amplitudes of vertical and torsional motion are 15 mm and 9° with 0 phase difference. The superposed aerodynamics match the force measured directly from the coupled motion, which demonstrates that the superposition assumption is still valid for the bridge deck in coupled motion. In order to quantitatively evaluate the superposition assumption, the predictive aerodynamic forces from superposing are compared with the forces directly measured from coupled motion by the coefficient of determination R^2, which is defined as

$$R^2 = 1 - \frac{\sum (F_{\text{se,coupled}} - F_{\text{se,superposed}})^2}{\sum F^2_{\text{se,coupled}}}. \tag{6}$$

When the aerodynamic forces from superposing is identical to the aerodynamic of coupled motion as in Figure 6, R^2 will be close to 1. This study performs the coupled motion tests for 3 different wind speeds (5, 7, 9 m/s) combining with

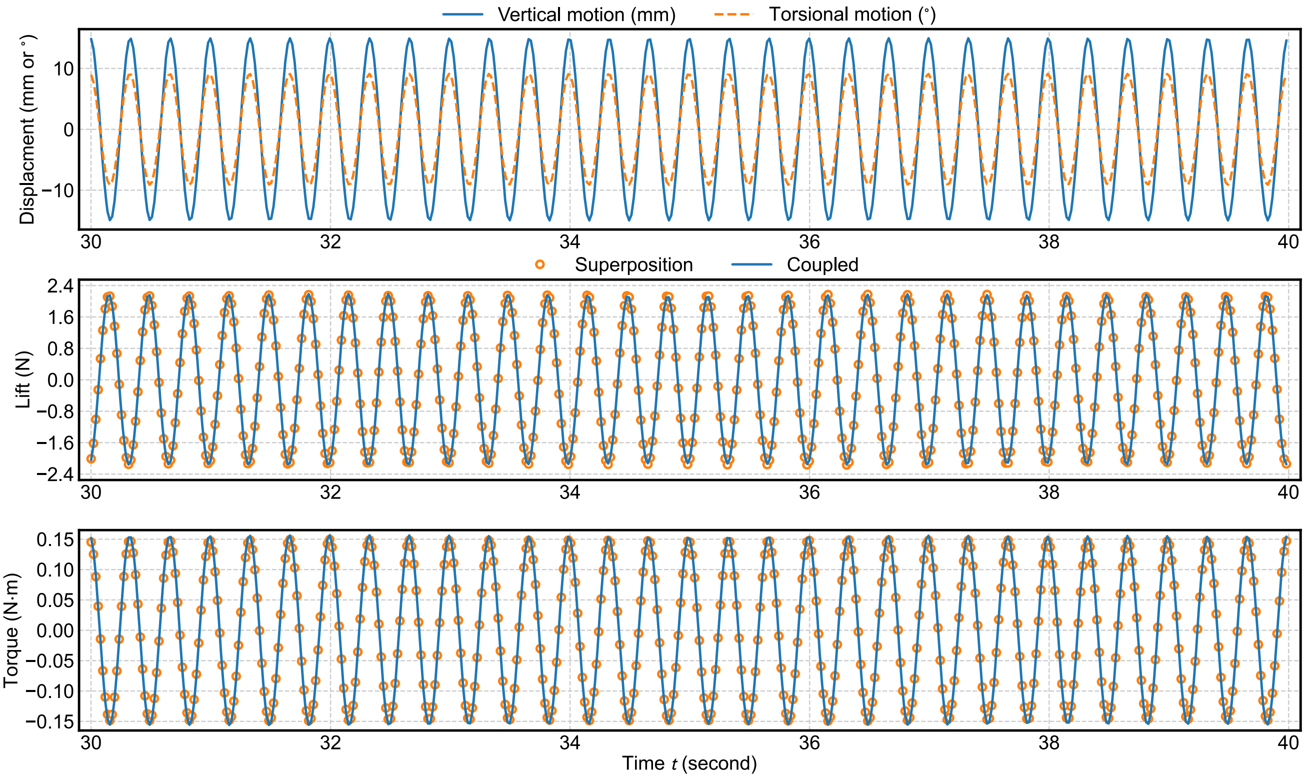


Fig. 6 Validation of superposition assumption for coupled vertical-torsional motion(Amplitudes of vertical and torsional motion are 15 mm and 9°, the wind speed is 9 m/s, and the phase difference is 0)

Table 2 Coupled forced-motion tests configurations

Wind Speed	Torsional Amplitude(°)	Vertical Amplitudes(mm)	Phase Difference (°)
5, 7, 9	3	5, 15	0,±16
5, 7, 9	5	7	0,±16
5, 7, 9	6	15	0
5, 7, 9	7	11	0,±16
5, 7, 9	9	13,15	0
5, 7, 9	12	13	0,±16

6 different torsional vibration amplitudes (3, 5, 6, 7, 9 and 12°). For each speed/amplitude combination, there are 3 or 4 implementation with different amplitudes ratios and phase differences. The whole coupled tests configuration is shown in Table 2.

Figure 7 plots the minimum $1 - R^2$ for vertical force and torque of different amplitudes ratio and phase difference for each combination of wind speeds and torsional amplitudes. Figure 7 clearly shows that the maximum $1 - R^2$ (or the minimum R^2) is smaller than 0.01 for all motion configurations. The similarity between superposed aerodynamic and directly measured aerodynamic from coupled motion proves superposition assumption with sufficient accuracy.

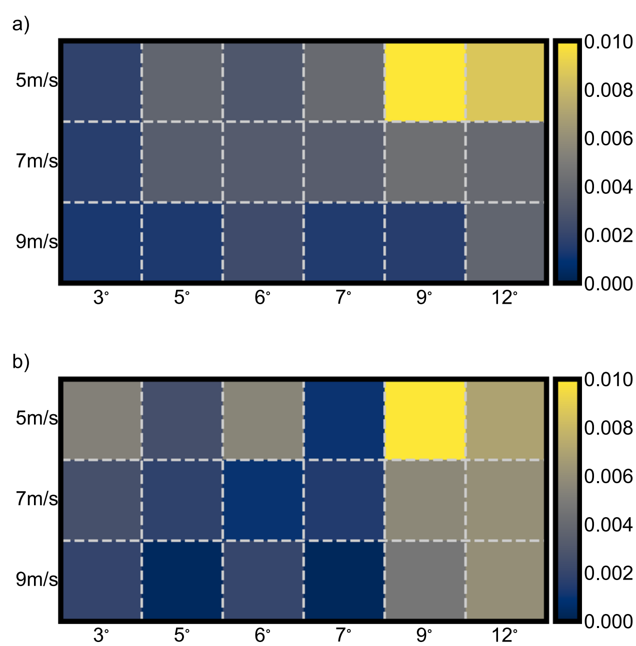


Fig. 7 Minimum of $1 - R^2$ for different combination of wind speeds and torsional amplitudes

The superposition assumption should be used with caution because it is only valid for specific sectional shape and

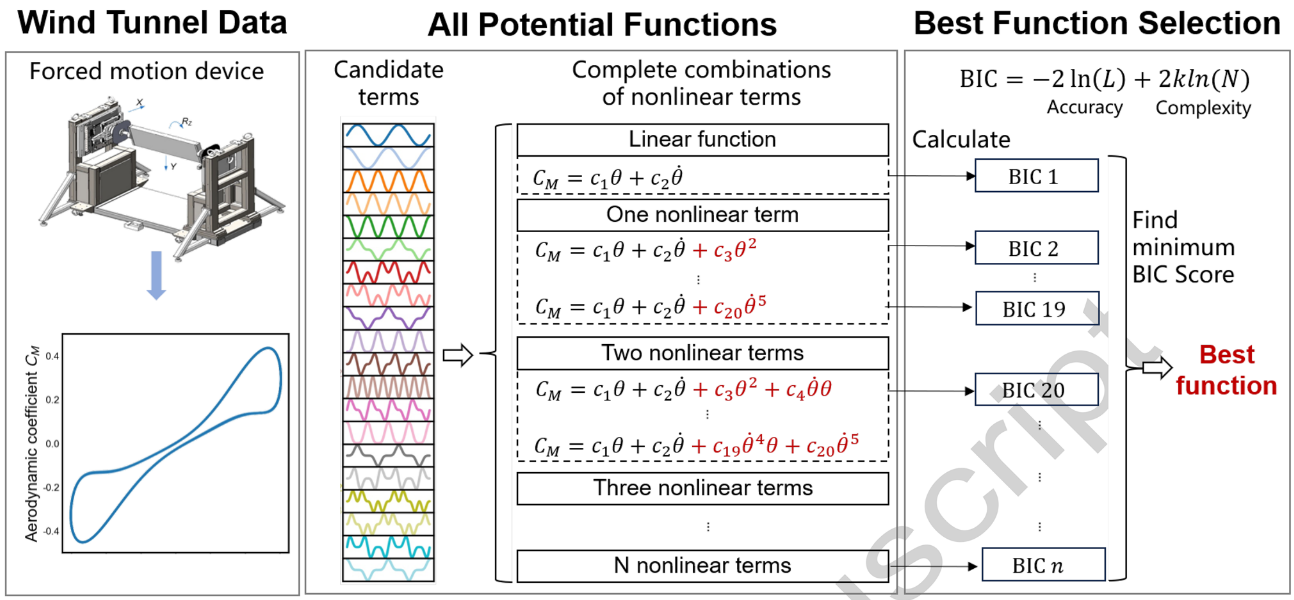
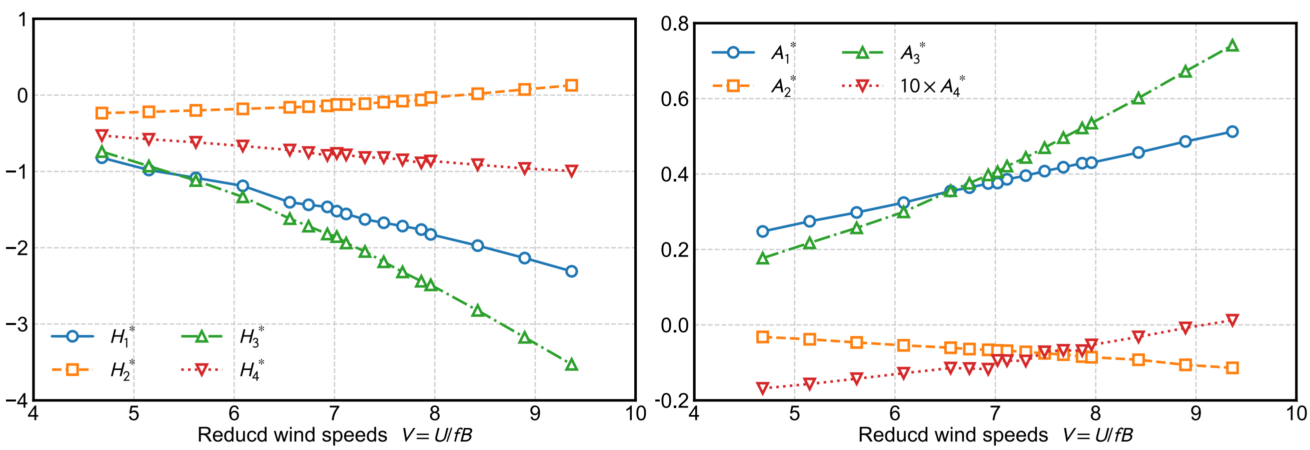


Fig. 8 The algorithm of nonlinear aeroelastic force inference from wind tunnel data: Aerodynamic force data is obtained from forced-motion wind tunnel test; the potential functions are listed by complete combinations with all candidate terms; Bayesian information criterion is used to find the optimal formula to represent the data



(a) H_1^*, H_2^*, H_3^* and H_4^*

(b) A_1^*, A_2^*, A_3^* and A_4^*

Fig. 9 Linear flutter derivatives of the bridge deck

within certain vibration amplitudes. If the vertical motion induced aerodynamics also contains nonlinear components, the superposition assumption is possible invalid.

2.4 Nonlinear aeroelastic force formula inference algorithm

To infer the nonlinear aeroelastic force form N_L, N_M, an optimal formula inference algorithm was proposed in Ma et al. [22]. It is used to find the simple and accurate semi-empirical aerodynamic force formula via experimental data. The algorithm diagram is shown in Figure 8.

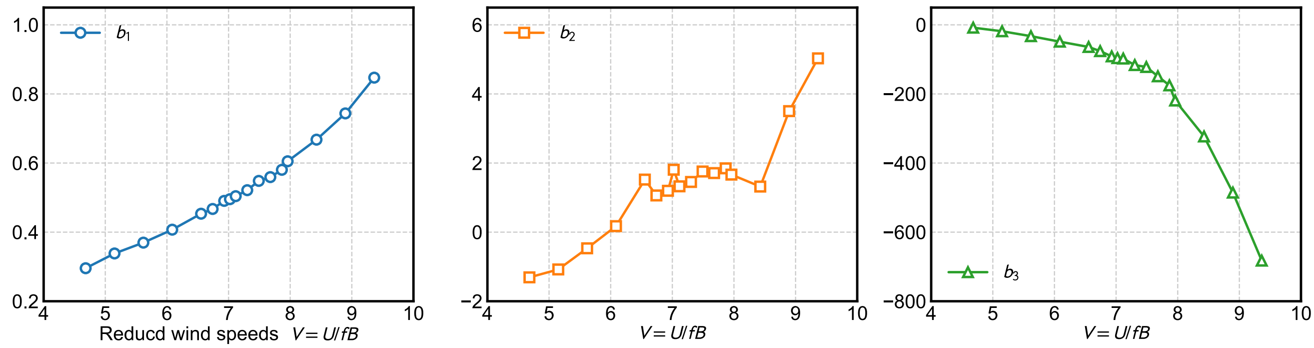
Taken $N_{M\theta}(\theta, \dot{\theta})$ as example, it can be expanded as n-degree polynomial form.

$$N_{M\theta} = \underbrace{c_3\theta^2 + c_4\theta\dot{\theta} + c_5\dot{\theta}^2}_{o^{(2)}} + o^{(3)} + o^{(4)} + o^{(5)} + o^n \tag{7}$$

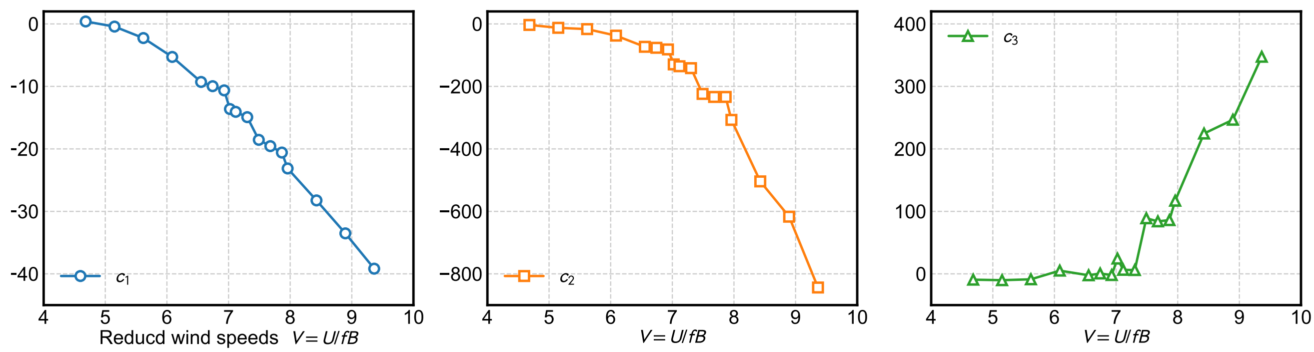
where c_i is the coefficient for each team, the superscript denotes the expansion order.

Based on the method in Ma et al. [22], the nonlinear aeroelastic force is identified as:

$$N_{yL}(y, \dot{y}) = 0 \tag{8a}$$



(a) b_1, b_2 and b_3



(b) c_1, c_2 and c_3

Fig. 10 Nonlinear aerodynamic coefficients of the bridge deck

$$N_{\theta L}(\theta, \dot{\theta}) = b_1 K^2 \theta \dot{\theta} + b_2 K^2 \theta \dot{\theta}^2 + b_3 K \dot{\theta}^5 \tag{8b}$$

$$N_{yM}(y, \dot{y}) = 0 \tag{8c}$$

$$N_{\theta M}(\theta, \dot{\theta}) = c_1 K^2 \theta \dot{\theta}^3 + c_2 K^3 \theta^2 \dot{\theta}^3 + c_3 K^4 \theta^3 \dot{\theta}^2 \tag{8d}$$

The vertical motion-induced aerodynamic lift and torque force only contain the linear part, and the nonlinear part is always 0, which is compatible with Figure 5.

Based on the algorithm in Figure 8, the linear and nonlinear aerodynamic coefficients can be identified at the same time. The linear part of the aerodynamic force can be expressed by the flutter derivatives as in Figure 9. Because A_4^* are small, its values are enlarged 10 times in Figure 9(b).

The nonlinear aerodynamic coefficients b_1, b_2, b_3 and c_1, c_2, c_3 in Eq.(8) are plotted in Figure 10. b_1 and b_2 varies gradually within the wind speeds interested in this study. b_2, b_3 and c_2, c_3 have two stages: When V is less than 8, their values vary slightly, but increase rapidly after $V > 8$. Especially for c_3, which is almost around 0 when $V < 8$, but increases very fast after $V > 8$.

Using the linear flutter derivatives in Figure 9 and the nonlinear coefficients in Figure 10, the aerodynamic induced by the vibration of the bridge deck can be reconstructed. Figure 11 plots the comparison of the measured aerodynamic from the forced-motion device and the calculated value using the formula derived from the inference algorithm used in this study, demonstrating that Eqs. (4) and (8) generally present linear aerodynamics at small amplitudes and nonlinear aerodynamics at large amplitudes. The aerodynamic modelling by Eq. (8) and parameters shown in Figures 9 and 10 are only suitable to the streamlined section shown in Figure 3, but the suitable aerodynamic modelling can be derived by the same method shown in this study.

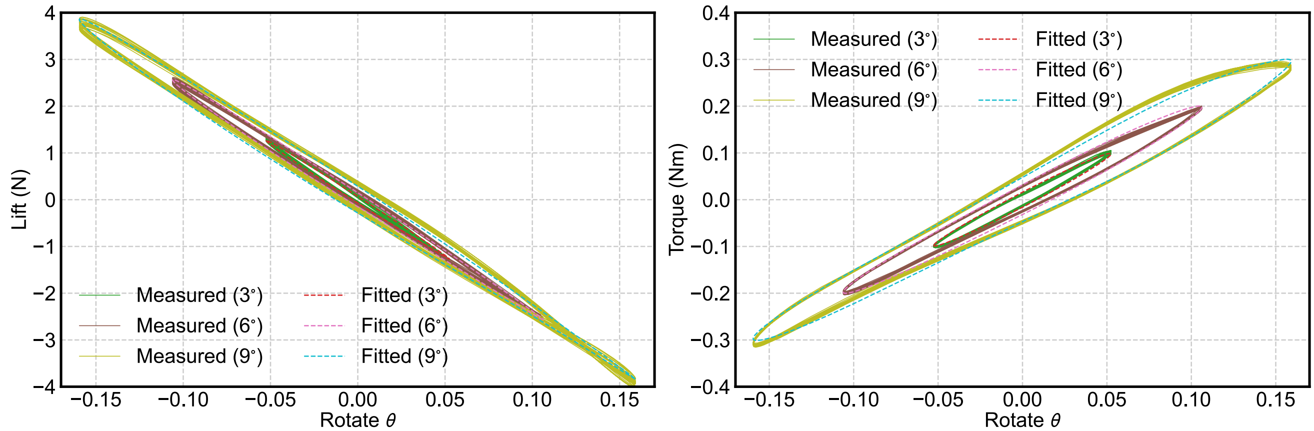
3 Nonlinear flutter response estimation by CxA method

3.1 Complexification of dynamics for flutter responses

Complexification of dynamics introduces a complex number to present both displacement and velocity [26]. In the flutter response, vertical and torsional motion can be represented as $\psi_y = \dot{y} + \iota K y$ and $\psi_\theta = \dot{\theta} + \iota K \theta$, where $\iota^2 = -1$ and K is the dimensionless vibration frequency. The conjugates of ψ_y and ψ_θ are $\dot{y} - \iota K y$ and $\dot{\theta} - \iota K \theta$, respectively.

Then the displacement, velocity and acceleration are

$$y = \frac{\psi_y - \psi_y^*}{2\iota K}, \quad \dot{y} = \frac{\psi_y + \psi_y^*}{2}, \quad \ddot{y} = \dot{\psi}_y - \iota K \frac{\psi_y + \psi_y^*}{2} \tag{9a}$$



(a) Aerodynamic Lift

(b) Aerodynamic Torque

Fig. 11 Comparison of measured aerodynamics and calculated force by Eqs. (4) and (8)

$$\theta = \frac{\psi_\theta - \psi_\theta^*}{2\iota K}, \quad \dot{\theta} = \frac{\psi_\theta + \psi_\theta^*}{2}, \quad \ddot{\theta} = \dot{\psi}_\theta - \iota K \frac{\psi_\theta + \psi_\theta^*}{2} \tag{9b}$$

Then the nonlinear flutter response is

$$\begin{aligned}
&\dot{\psi}_y - \iota K \frac{\psi_y + \psi_y^*}{2} + 2\xi_y K_y \frac{\psi_y + \psi_y^*}{2} + K_y^2 \frac{\psi_y - \psi_y^*}{2\iota K} \\
&= m_y^* \Bigg[K H_1^* \frac{\psi_y + \psi_y^*}{2} + K H_2^* \frac{\psi_\theta + \psi_\theta^*}{2} + K^2 H_3^* \frac{\psi_\theta - \psi_\theta^*}{2\iota K} \\
&+ K^2 H_4^* \frac{\psi_y - \psi_y^*}{2\iota K} + b_1 K^2 \frac{\psi_\theta - \psi_\theta^*}{2\iota K} \frac{\psi_\theta + \psi_\theta^*}{2} \\
&+ b_2 K^2 \frac{\psi_\theta - \psi_\theta^*}{2\iota K} \left(\frac{\psi_\theta + \psi_\theta^*}{2}\right)^2 + b_3 K \left(\frac{\psi_\theta + \psi_\theta^*}{2}\right)^5 \Bigg]
\end{aligned} \tag{10a}$$

$$\begin{aligned}
&\dot{\psi}_\theta - \iota K \frac{\psi_\theta + \psi_\theta^*}{2} + 2\xi_\theta K_\theta \frac{\psi_\theta + \psi_\theta^*}{2} + K_\theta^2 \frac{\psi_\theta - \psi_\theta^*}{2\iota K} \\
&= m_\theta^* \Bigg[K A_1^* \frac{\psi_y + \psi_y^*}{2} + K A_2^* \frac{\psi_\theta + \psi_\theta^*}{2} + K^2 A_3^* \frac{\psi_\theta - \psi_\theta^*}{2\iota K} \\
&+ K^2 A_4^* \frac{\psi_y - \psi_y^*}{2\iota K} + c_1 K^2 \frac{\psi_\theta - \psi_\theta^*}{2\iota K} \left(\frac{\psi_\theta + \psi_\theta^*}{2}\right)^3 \\
&+ c_2 K^3 \left(\frac{\psi_\theta - \psi_\theta^*}{2\iota K}\right)^2 \left(\frac{\psi_\theta + \psi_\theta^*}{2}\right)^3 \\
&+ c_3 K^4 \left(\frac{\psi_\theta - \psi_\theta^*}{2\iota K}\right)^3 \left(\frac{\psi_\theta + \psi_\theta^*}{2}\right)^2 \Bigg]
\end{aligned} \tag{10b}$$

3.2 Flutter response evolutionary function by CxA method

The dynamics of $\psi_y = \phi_y e^{\iota K\tau}$ and $\psi_\theta = \phi_\theta e^{\iota K\tau}$ are decomposed into slow, ϕ_y and ϕ_θ, and fast, $e^{\iota K\tau}$, components. The slow components ϕ_y and ϕ_θ represent the evolution of amplitudes and phase, while the fast $e^{\iota K\tau}$ just simply represents the sinusoidal vibration cycle by cycle. The conjugates of ψ_y and ψ_θ are $\psi_y^* = \phi_y^* e^{-\iota K\tau}$ and $\psi_\theta^* = \phi_\theta^* e^{-\iota K\tau}$. Therefore, the averaging method can extract the slow components by averaging out the fast components. Complexification is suitable for nonlinear dynamics because it converts complicated high-order trigonometric functions into straightforward complex algebra operation.

Substituting $\psi_y = \phi_y e^{\iota K\tau}$ and $\psi_\theta = \phi_\theta e^{\iota K\tau}$ into Eq. (10) and averaging out the fast components, the flutter response becomes

$$\begin{aligned}
&\dot{\phi}_y + \iota K \frac{\phi_y}{2} + \xi_y K_y \phi_y - \iota K_y^2 \frac{\phi_y}{2K} \\
&= m_y^* K \Bigg[H_1^* \frac{\phi_y}{2} - \iota H_4^* \frac{\phi_y}{2} + H_2^* \frac{\phi_\theta}{2} - \iota H_3^* \frac{\phi_\theta}{2} \\
&- \iota b_2 \frac{\phi_\theta^2 \phi_\theta^*}{8} + b_3 \frac{5\phi_\theta^3 \phi_\theta^{*2}}{16} \Bigg]
\end{aligned} \tag{11a}$$

$$\begin{aligned}
&\dot{\phi}_\theta + \iota K \frac{\phi_\theta}{2} + \xi_\theta K_\theta \phi_\theta - \iota K_\theta^2 \frac{\phi_\theta}{2K} \\
&= m_\theta^* K \Bigg[A_1^* \frac{\phi_y}{2} + A_2^* \frac{\phi_\theta}{2} - \iota A_3^* \frac{\phi_\theta}{2} - \iota A_4^* \frac{\phi_y}{2} \\
&+ c_2 \frac{\phi_\theta^3 \phi_\theta^{*2}}{16} - \iota c_3 \frac{\phi_\theta^3 \phi_\theta^{*2}}{16} \Bigg]
\end{aligned} \tag{11b}$$

Next, ϕ_y and ϕ_θ can be further decomposed into polar form as $\phi_y = A_y e^{\iota\beta_y}$ and $\phi_\theta = A_\theta e^{\iota\beta_\theta}$, where A_y, A_θ are the amplitudes of vertical and torsional vibration, and β_y, β_θ are phases.

The evolutionary functions of A_y, A_θ and β_y, β_θ are

$$\begin{aligned}
&\dot{A}_y + \iota\dot{\beta}_y A_y + \iota K \frac{A_y}{2} + \xi_y K_y A_y - \iota K_y^2 \frac{A_y}{2K} \\
&= m_y^* K \frac{A_y}{2} \left(H_1^* - \iota H_4^*\right) + m_y^* K \frac{A_\theta e^{\iota(\beta_\theta - \beta_y)}}{2} \left(H_2^* - \iota H_3^*\right)
\end{aligned}$$

$$-\iota m_y^* K b_2 \frac{A_\theta^3 e^{\iota(\beta_\theta-\beta_y)}}{8} + m_y^* K b_3 \frac{5A_\theta^5 e^{\iota(\beta_\theta-\beta_y)}}{16} \tag{12a}$$

$$\begin{aligned}&\dot{A}_\theta + \iota\dot{\beta}_\theta A_\theta + \iota K\frac{A_\theta}{2} + \xi_\theta K_\theta A_\theta - \iota K_\theta^2 \frac{A_\theta}{2K} \\ &= m_\theta^* K A_1^* \frac{A_y}{2} e^{\iota(\beta_y-\beta_\theta)} - \iota m_\theta^* K A_4^* \frac{A_y}{2} e^{\iota(\beta_y-\beta_\theta)} \\ &\quad + m_\theta^* K \left[A_2^* \frac{A_\theta}{2} + c_2 \frac{A_\theta^5}{16}\right] - \iota m_\theta^* K \left[A_3^* \frac{A_\theta}{2} + c_3 \frac{A_\theta^5}{16}\right]\end{aligned} \tag{12b}$$

The complex phase can be expanded based on Euler's formula

$$e^{\iota(\beta_\theta-\beta_y)} = \cos(\beta_\theta-\beta_y) + \iota\sin(\beta_\theta-\beta_y) \tag{13a}$$

$$e^{\iota(\beta_y-\beta_\theta)} = \cos(\beta_y-\beta_\theta) + \iota\sin(\beta_y-\beta_\theta) \tag{13b}$$

The real and imaginary parts of Eq. (12) can be decoupled as amplitude evolution equations in Eq. (14) and phase evolution equation in Eq. (15).

$$\begin{aligned}&\dot{A}_y + \xi_y K_y A_y = m_y^* K \left[\frac{A_y}{2} H_1^* \right. \\ &\quad + \frac{A_\theta}{2} H_2^* \cos(\beta_\theta-\beta_y) + \frac{A_\theta}{2} H_3^* \sin(\beta_\theta-\beta_y) \\ &\quad \left. + b_2 \frac{A_\theta^3}{8} \sin(\beta_\theta-\beta_y) + b_3 \frac{5A_\theta^5}{16}\cos(\beta_\theta-\beta_y)\right]\end{aligned} \tag{14a}$$

$$\begin{aligned}&\dot{A}_\theta + \xi_\theta K_\theta A_\theta = m_\theta^* K \left[A_1^* \frac{A_y}{2}\cos(\beta_y-\beta_\theta) \right. \\ &\quad \left. + A_4^* \frac{A_y}{2}\sin(\beta_y-\beta_\theta) + A_2^* \frac{A_\theta}{2} + c_2 \frac{A_\theta^5}{16}\right]\end{aligned} \tag{14b}$$

$$\begin{aligned}&\dot{\beta}_y A_y + K\frac{A_y}{2} - K_y^2 \frac{A_y}{2K} = m_y^* K \left[-\frac{A_y}{2} H_4^* \right. \\ &\quad - \frac{A_\theta}{2} H_3^* \cos(\beta_\theta-\beta_y) + \frac{A_\theta}{2} H_2^* \sin(\beta_\theta-\beta_y) \\ &\quad \left. - b_2 \frac{A_\theta^3}{8}\cos(\beta_\theta-\beta_y) + b_3 \frac{5A_\theta^5}{16}\sin(\beta_\theta-\beta_y)\right]\end{aligned} \tag{15a}$$

$$\begin{aligned}&\dot{\beta}_\theta A_\theta + K\frac{A_\theta}{2} - K_\theta^2 \frac{A_\theta}{2K} = m_\theta^* K \left[A_1^* \frac{A_y}{2}\sin(\beta_y-\beta_\theta) \right. \\ &\quad \left. - A_4^* \frac{A_y}{2}\cos(\beta_y-\beta_\theta) - A_3^* \frac{A_\theta}{2} - c_3 \frac{A_\theta^5}{16}\right]\end{aligned} \tag{15b}$$

Let $A_\theta \times$Eq. (15a) $- A_y \times$Eq. (15b) and $\beta = \beta_y - \beta_\theta$, the final nonlinear flutter evolution equation is

$$\begin{aligned}\dot{A}_y &= m_y^* K \left[\frac{A_y}{2} H_1^* + \frac{A_\theta}{2} H_2^* \cos\beta - \frac{A_\theta}{2} H_3^* \sin\beta \right. \\ &\quad \left. - b_2 \frac{A_\theta^3}{8}\sin\beta + b_3 \frac{5A_\theta^5}{16}\cos\beta\right] - \xi_y K_y A_y\end{aligned} \tag{16a}$$

$$\begin{aligned}\dot{A}_\theta &= m_\theta^* K \left[A_1^* \frac{A_y}{2}\cos\beta + A_4^* \frac{A_y}{2}\sin\beta + A_2^* \frac{A_\theta}{2} \right. \\ &\quad \left. + c_2 \frac{A_\theta^5}{16}\right] - \xi_\theta K_\theta A_\theta\end{aligned} \tag{16b}$$

$$\begin{aligned}\dot{\beta} &= m_y^* K \left[-\frac{1}{2} H_4^* - \frac{A_\theta}{2A_y} H_3^* \cos\beta - \frac{A_\theta}{2A_y} H_2^* \sin\beta \right. \\ &\quad \left. - b_2 \frac{A_\theta^3}{8A_y}\cos\beta + b_3 \frac{5A_\theta^5}{16A_y}\sin\beta\right] \\ &\quad - m_\theta^* K \left[A_1^* \frac{A_y}{2A_\theta}\sin\beta - A_4^* \frac{A_y}{2A_\theta}\cos\beta - A_3^* \frac{1}{2} - c_3 \frac{A_\theta^4}{16}\right] \\ &\quad + \frac{K_y^2 - K_\theta^2}{2K}\end{aligned} \tag{16c}$$

In order to solve the above differential equation, several initial values should be determined first, including the vibration frequency K, phase difference β and two amplitudes A_y and A_θ. Throughout the nonlinear flutter response from initial condition to stable vibration, the vibration frequency K is assumed as constant, which is demonstrated by other experiments [7, 27]. The other reason of constant vibration frequency K is because there is not nonlinear aerodynamic stiffness term in Eq. (8).

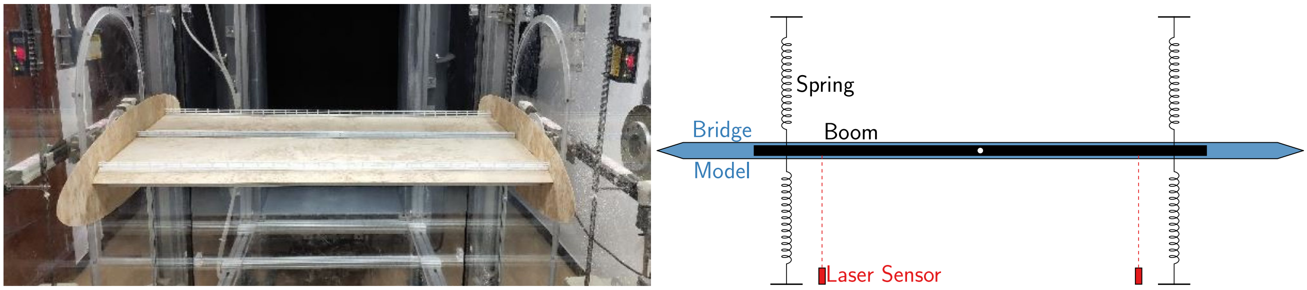
3.3 Determination of initial condition based on complex eigenvalues

When determining the initial condition of the vibration of flutter, the nonlinear aerodynamic is not obvious, and it can be calculated by the linear part of Eq. (4). The flutter analysis based on linear derivatives was well studied 20 years ago, and the most popular method is complex eigenvalues [28, 29]. Eq. (4) can be rewritten as first-order dynamic equation

$$d\begin{Bmatrix} y \\ \theta \\ \dot{y} \\ \dot{\theta}\end{Bmatrix}/d\tau = \begin{bmatrix} \multicolumn{2}{c}{\mathbf{0}} & \multicolumn{2}{c}{\boldsymbol{I}} \\ m_y^* H_4^* K^2 - K_y^2 & m_y^* H_3^* K^2 & m_y^* H_1^* K - 2\xi_y K_y & m_y^* H_2^* K \\ m_\theta^* A_4^* K^2 & m_\theta^* A_3^* K^2 - K_\theta^2 & m_\theta^* A_1^* K & m_\theta^* A_2^* K - \xi_\theta K_\theta \end{bmatrix} \begin{Bmatrix} y \\ \theta \\ \dot{y} \\ \dot{\theta}\end{Bmatrix} \tag{17}$$

The real part of the eigenvalue for the above system matrix is the damping of the system $\bar{\xi}$, and the imaginary part is the frequency of the system $\bar{K}$. For different wind speeds U, the flutter frequency K and other parameters for the initial condition is summarized concisely as following step and more details can be found in Ding et al. [28] and Starossek [29]:

1. increase U from 0
2. iterate K from K_y to K_θ



(a) Sectional model in the TJ-7 wind tunnel (b) Schematic plot of the sectional model)

Fig. 12 Suspended Bridge deck sectional model tested in wind tunnel

3. substitute $H_1^*(K)$, $H_2^*(K)$, $H_3^*(K)$, $H_4^*(K)$ $A_1^*(K)$, $A_2^*(K)$, $A_3^*(K)$, $A_4^*(K)$ into Eq. (17), and calculate complex modal frequency $\bar{K}$
4. calculate the complex modal damping $\bar{\xi}$ as long as $\bar{K} = K$

When $\bar{\xi} < 0$, the system is stable and flutter will not occur; When $\bar{\xi} = 0$, the corresponding wind speed is the critical wind speed U_{cr}. For $U > U_{\mathrm{cr}}$, the flutter frequency will be K if system frequency $\bar{K}$ is the same as the frequency used to determine the derivatives of the flutter. The phase β and the amplitude ratio A_y/A_θ can be calculated from the eigenvector subsequently.

3.4 Flutter response evolutionary function for nonlinear structural damping

The above formulas are based on linear structural damping. However, nonlinear structural damping exists widely in real structures and suspension systems for bridge deck wind tunnel tests [30]. Nonlinear structural damping can also be expanded by Taylor series. If the expansion order is 2, the structural damping is

$$\xi_y(y) = \xi_{y0} + \xi_{y1}y + \xi_{y2}y^2 \tag{18a}$$
$$\xi_\theta(\theta) = \xi_{\theta 0} + \xi_{\theta 1}\theta + \xi_{\theta 2}\theta^2 \tag{18b}$$

Based on the same CxA procedure, the nonlinear flutter evolutions equation with nonlinear structural damping is

$$\begin{aligned}\dot{A}_y = {} & m_y^* K\left[\frac{A_y}{2}H_1^* + \frac{A_\theta}{2}H_2^*\cos\beta - \frac{A_\theta}{2}H_3^*\sin\beta\right.\\ & \left.-b_2\frac{A_\theta^3}{8}\sin\beta + b_3\frac{5A_\theta^5}{16}\cos\beta\right] - \left(\xi_{y0}K_yA_y + \xi_{y2}\frac{A_y^3}{4K_y}\right)\end{aligned} \tag{19a}$$

$$\begin{aligned}\dot{A}_\theta = {} & m_\theta^* K\left[A_1^*\frac{A_y}{2}\cos\beta + A_4^*\frac{A_y}{2}\sin\beta + A_2^*\frac{A_\theta}{2} + c_2\frac{A_\theta^5}{16}\right]\\ & - \left(\xi_{\theta 0}K_\theta A_\theta + \xi_{\theta 2}\frac{A_\theta^3}{4K_\theta}\right)\end{aligned} \tag{19b}$$

$$\begin{aligned}\dot{\beta} = {} & m_y^* K\left[-\frac{1}{2}H_4^* - \frac{A_\theta}{2A_y}H_3^*\cos\beta - \frac{A_\theta}{2A_y}H_2^*\sin\beta\right.\\ & \left.-b_2\frac{A_\theta^3}{8A_y}\cos\beta + b_3\frac{5A_\theta^5}{16A_y}\sin\beta\right]\\ & -m_\theta^* K\left[A_1^*\frac{A_y}{2A_\theta}\sin\beta - A_4^*\frac{A_y}{2A_\theta}\cos\beta - A_3^*\frac{1}{2} - c_3\frac{A_\theta^4}{16}\right]\\ & + \frac{K_y^2 - K_\theta^2}{2K}\end{aligned} \tag{19c}$$

Table 3 Dynamic characteristics of suspension system

Vertical		Torsional	
m	2.01 kg	I	4.26×10^{-2} kg·m
f_y	2.26 Hz	f_θ	3.12 Hz

4 Nonlinear flutter response estimation

4.1 Suspension system of bridge deck and its mechanical damping

The free vibration tests are performed to validate the response estimation algorithm proposed in this study. The sectional model and its suspension system with one boom and four springs are illustrated in Figure 12.

The dynamic characteristics of the suspension system are listed in Table 3. The free decay vibration tests without wind speed is performed to determine the mechanical damping of suspension system. Five repeat free decay vibrations were tested in the vertical and torsional directions. The amplitude-dependent damping identification algorithm in Gao and Zhu [30] is employed, and the results are plotted in Figure 13. When the amplitudes are small, the damping variation is very complex. Therefore, only damping with large amplitudes is

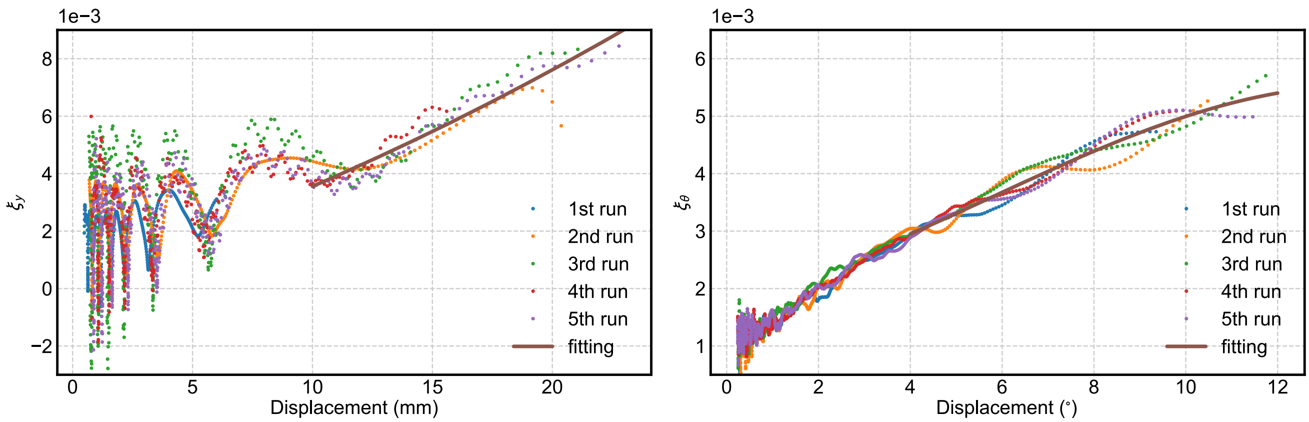
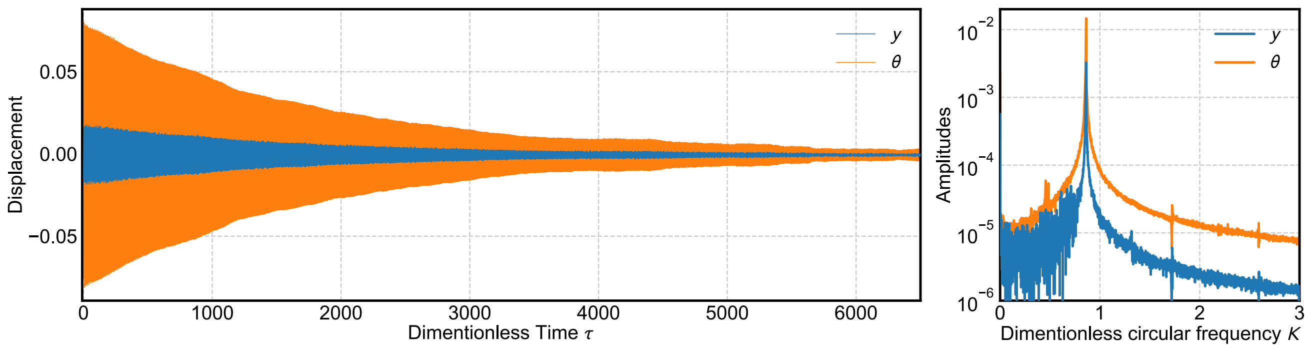
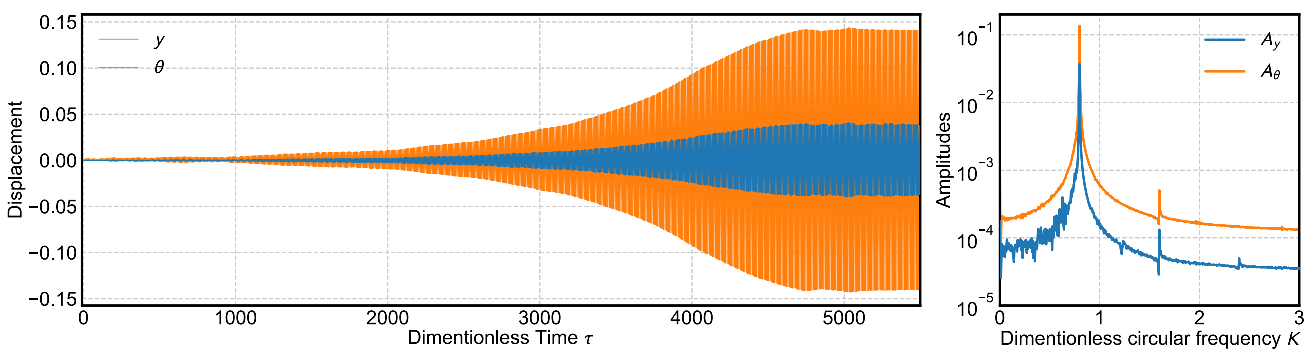


Fig. 13 Amplitude dependent mechanical damping of sectional model



(a) $U = 7.7$ m/s



(b) $U = 8.2$ m/s

Fig. 14 Suspended Bridge deck sectional model tested in wind tunnel

used to fit the nonlinear mechanical damping formulas in Eq. (18), and the fitting results are shown in Figure 13.

4.2 Nonlinear flutter amplitudes from free vibration

Free vibrations were performed at different wind speeds to test the flutter responses of the bridge section. Two typical test results with $U = 7.7$ m/s and 8.2m/s are plotted in Figure 14. Figure 14(a) shows that the amplitudes decay to 0. $U = 7.7$ m/s is a little smaller than the critical wind speed, and the vertical and torsional motion are almost coupled. $U = 8.2$ m/s is greater than the critical speed, and Figure 14b shows the amplitudes that grow from 0 to the limit-cycle oscillation (LCO).

Repeating the free vibration at different wind speeds, Figure 15 shows the stable LCO amplitudes, which will be used as targets for estimating the flutter response.

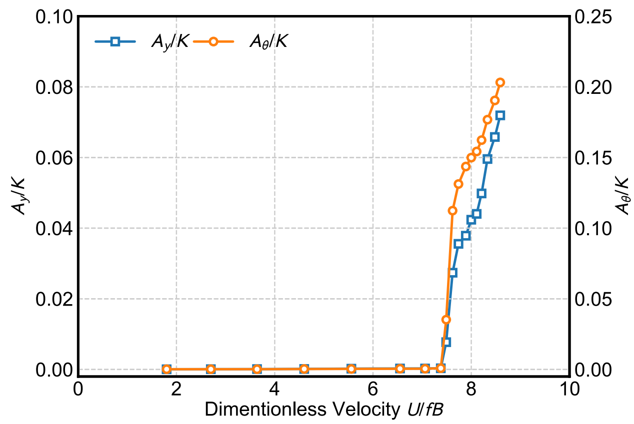


Fig. 15 Vertical and torsional amplitudes of limit-cycle oscillation for different wind speeds

4.3 Flutter responses estimation and validation with free-vibration testing results

With the dynamic characteristics of the sectional model in Table 3 and the nonlinear aerodynamic coefficients in Figures 9 and 10, the amplitudes of the flutter response and phase difference can be evaluated by Eq. (16) or Eq. (19) as long as the initial conditions are given. At the initial stage, the vibration frequencies K, phase difference β and amplitudes ratio A_y/A_θ at $\tau = 0$ is derived by the complex eigenvalue mode analysis of Eq. (17). It should be noted that vibration frequencies K are kept constant from initial vibration until LCO is reached, because flutter is a fluid-structure interaction procedure, and once flutter is excited, the fluid field and structure are mutually interacted at a fixed frequency K.

Figure 16 plots the flutter response of the sectional model when $U = 8.0$ m/s and 8.5 m/s. The vibration data are derived by time integration of Eq. (4) using Runge-Kutta methods, and the amplitudes A_y and A_θ are evaluated by Eq. (19). Assuming that $A_\theta = 0.01$ at $\tau = 0$, y, θ, A_y, and A_θ are calculated subsequently, and the exact match between y and A_y, as well as θ and A_θ, proves the precision of the amplitude evaluation function in Eq. (19). In addition to amplitudes, the phase difference β between vertical and torsional motion is also plotted in Figure 17, which shows that the phase difference will also change as the amplitudes evolve. For higher wind speeds, β changes more than for lower wind speeds.

When vibration enters the LCO stage, the differential of amplitudes and phase difference are all zeros, then the stable amplitudes and phase difference can be solved by Eq. (19) as multivariate equations. The calculated amplitudes of Eq. (19) are compared with the free vibration testing results in Figure 15, and the comparison results are plotted in Figure 17. Using equivalent linear mechanical damping, the flutter responses can be estimated by Eq. (16), which is also plotted in Figure 17 as a reference.

The estimation of the flutter response with nonlinear mechanical damping is in very good agreement with the test results. However, at large torsional amplitudes, the estimation results are smaller than the tests, because the vertical spring cannot remain vertical, which will affect the stiffness of the suspension system for the bridge section [31]. The estimation with linear mechanical stiffness has a smaller discrepancy.

5 Response evolution characteristics of nonlinear flutter

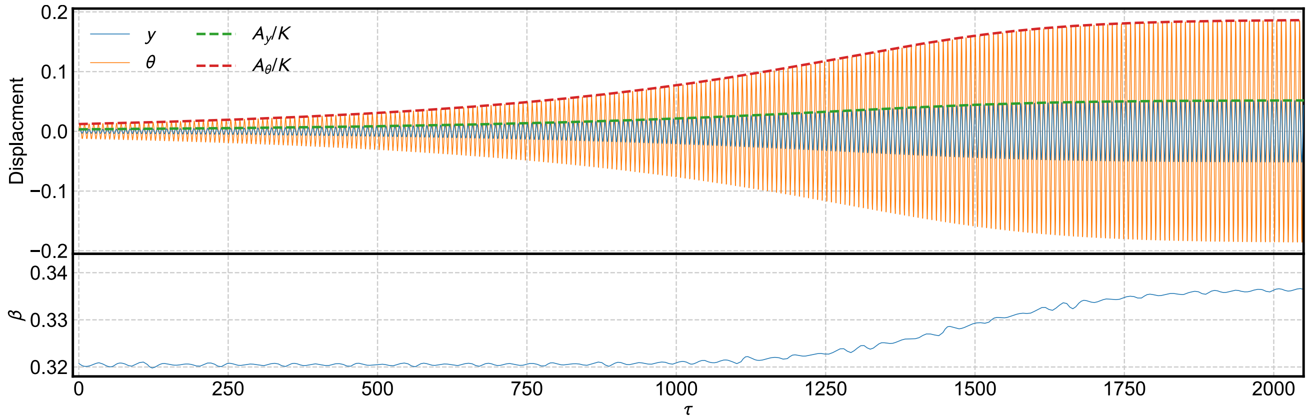
Based on the differential equation of amplitude Eq. (19), the evolution at different amplitudes can be evaluated. Figure 18 plots the streamline of two amplitudes assuming a constant phase difference β, whose value is taken when the LCO is stable. Each streamline in Figure 18 means the envelope of amplitude evolution from different initial states.

Figure 18 demonstrates that there is straight line from 0 to the stable LCO amplitudes. It means, from any initial condition, flutter response will first approaches the straight line first, then developes to the stable LCO amplitudes. This "straight line" in Figure 18 means that the ratio between vertical and torsional amplitudes is constant from initial flutter until stable LCO. It should be noted that the constant amplitudes ratio is possibly dependent on the linear vertical motion induced aerodynamics. If the vertical motion induced force is also nonlinear, the responses is more complex.

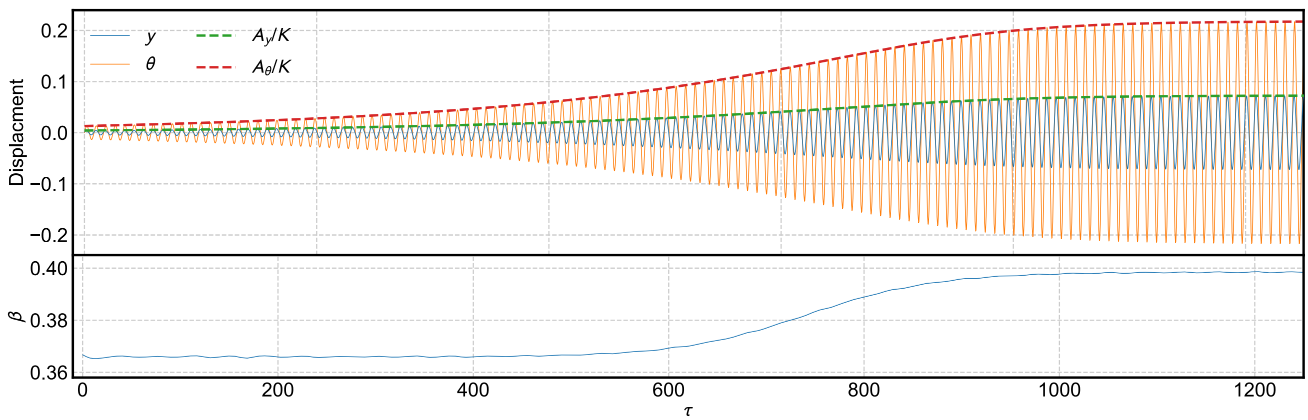
Due to the constant relationship between the vertical and the torsional amplitudes, the evolution of the phase difference β can only be investigated with the torsional amplitude A_θ as the 2D streamline plot in Figure 19. For $U = 8.0$ m/s, the changes of β are insignificant. However, for $U = 8.5$ m/s, as the amplitudes increase, β increases around 0.07 radians (4°). That means, for large flutter response amplitudes, the greater nonlinearity in the aerodynamic force will change the phase difference between vertical and torsional motion. If β is assumed to be constant from initial flutter to stable LCO, errors will be introduced into the amplitude estimation.

6 Conclusions

In this study, a nonlinear flutter response of long-span bridges estimation is established using the aerodynamic data measured by force-motion equipment. This method has two parts: a nonlinear aerodynamic model and a response prediction. The nonlinear aerodynamic force data was measured by forced-motion equipment with different wind speeds and amplitudes. Traditional flutter derivatives are used to model the linear part of aerodynamic force, and, for the nonlinear part, the autonomous inference algorithm is employed to establish and identify a concise and accurate polynomial for-



(a) $U = 8.0$ m/s



(b) $U = 8.5$ m/s

Fig. 16 Suspended Bridge deck sectional model tested in wind tunnel

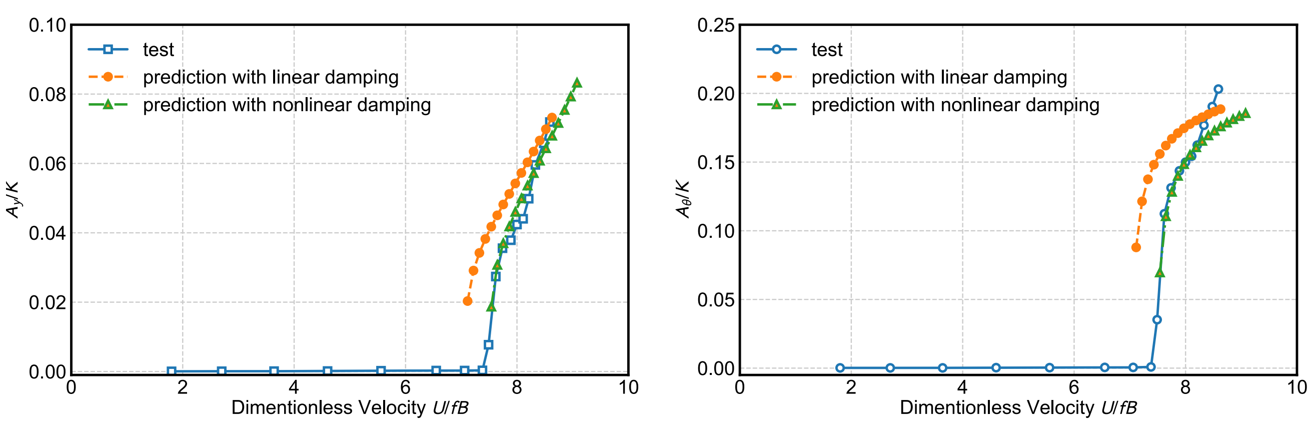


Fig. 17 Vertical and torsional amplitudes of limit-cycle oscillation for different wind speeds

mula. Overall, for each wind speed (or reduced frequency), there are 14 coefficients total in the nonlinear aerodynamic model composed of 8 flutter derivatives and 6 nonlinear parameters. Based on the established aerodynamic model, the complexification-averaging method is used to derive the differential equations of the response amplitudes and the phase difference. The amplitudes of flutter in a stable LCO stage can be estimated when the differential equations are zero. The comparison between the estimated response and the experimental results of the sectional model demonstrates the precision of the proposed method. On the basis of the amplitude differential equations, it is also found that the coupled vertical-torsional flutter tends to have a fixed amplitude ratio from initial flutter to stable LCO stage. However, the

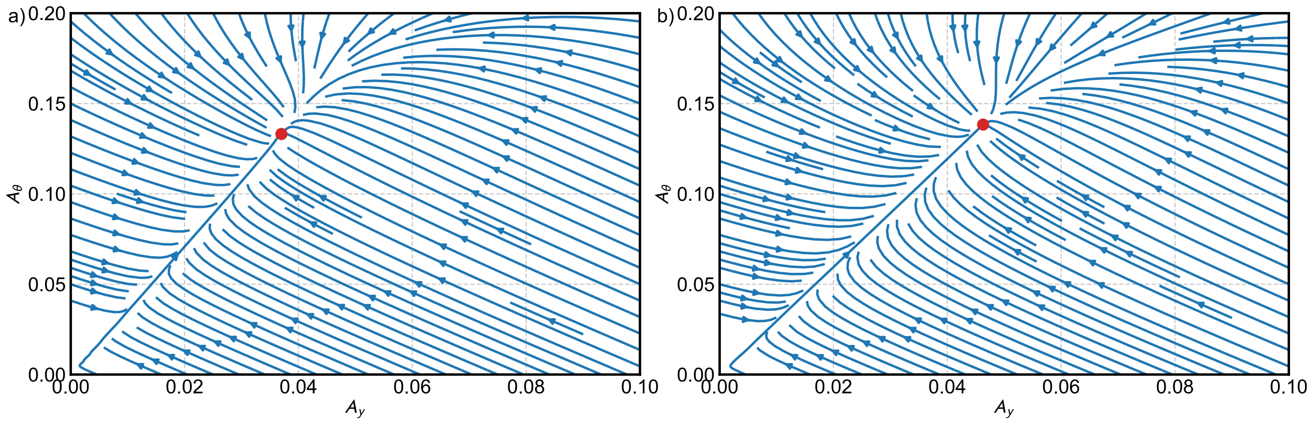


Fig. 18 Streamline of vertical and torsional amplitudes (legends: ●: stable LCO amplitudes. a: $U = 8.0$ m/s; b: $U = 8.5$ m/s)

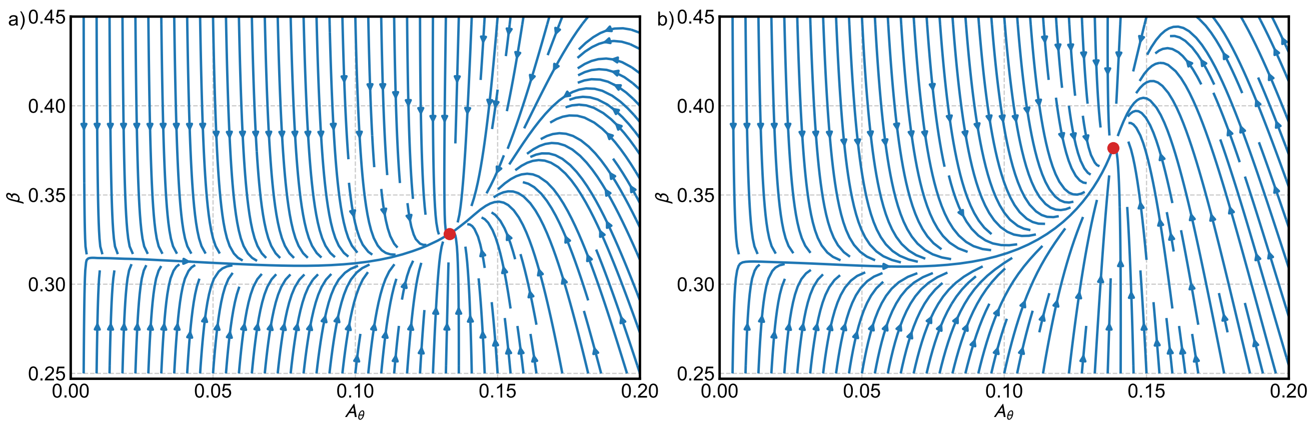


Fig. 19 Streamline of torsional amplitudes and phase difference (legends: ●: stable LCO amplitudes. a: $U = 8.0$ m/s; b: $U = 8.5$ m/s)

phase difference between vertical-torsional motion changes as the amplitudes increase. It may introduce errors if the phase difference is assumed as constant.

Author Contributions The conception and design of the study was a collaborative effort between all authors. Wei Cui was responsible for the conception, modeling and computation. Wei Cui wrote the first draft of the manuscript, and Lin Zhao supervised the entire study. Both Wei Cui and Lin Zhao provided financial support for the research. All authors read and approved the final version of the manuscript.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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